MATHEMATICS

Analysis and approaches

Higher Level | Set A | Paper 1

Mock examination · 2025 series

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

- A calculator is not permitted.
- This paper has Section A and Section B. Both sections are compulsory.
- Answer every question. Show all working and give sufficient mathematical reasoning.
- A clean Mathematics: analysis and approaches formula booklet may be used.
- Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.
- Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.
- The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 5]

The function f is defined by $f(x) = \frac{x-1}{2-x}$, for $x \in \mathbb{R} \setminus \{2\}$.

(a) State the range of f. [1]

(b) Find $f^{-1}(x)$. [4]

SECTION A

Question 2 [Maximum mark: 6]

All angles in this question are in radians.

Find the exact value of $\int_0^{\pi/2} 8 \sin x \cos x \, \mathrm{d}x$. [6]

SECTION A

Question 3 [Maximum mark: 6]

Let $f(x) = ax^3 + 2ax^2 + bx + 3a$, where $a, b \in \mathbb{R}$ and $a \neq 0$. Its three zeros are counted with multiplicity.

(a) Write down the sum and the product of the zeros of f. [2]

(b) The curve $y = f(x)$ has a stationary point at $(1, 3)$. Find a and b. [4]

SECTION A

Question 4 [Maximum mark: 5]

The discrete random variable X has the following probability distribution, where $0 \leq p \leq 0.7$ and $q \geq 0$.

x	1	2	3
$\mathrm{P}(X = x)$	p	$0.7 - p$	q

(a) Find q. [2]

(b) Given that $\mathrm{E}(X) = 2.1$, find p. [3]

SECTION A

Question 5 [Maximum mark: 6]

Solve $3\log_8(x+2) - 2\log_2(x+1) = 1$, giving an exact answer. [6]

SECTION A

Question 6 [Maximum mark: 6]

An arithmetic sequence has first term u_1 and common difference d. Let S_n denote the sum of its first n terms. It is given that $S_3 = S_8$.

(a) Show that $u_1 = -5d$. [4]

(b) Find the sixth term of the sequence. [2]

SECTION A

Question 7 [Maximum mark: 6]

Angles are measured in degrees.

Solve $\sin x + \cos x = \sqrt{\frac{3}{2}}$ for $0° < x < 90°$, giving exact values. [6]

SECTION A

Question 8 [Maximum mark: 6]

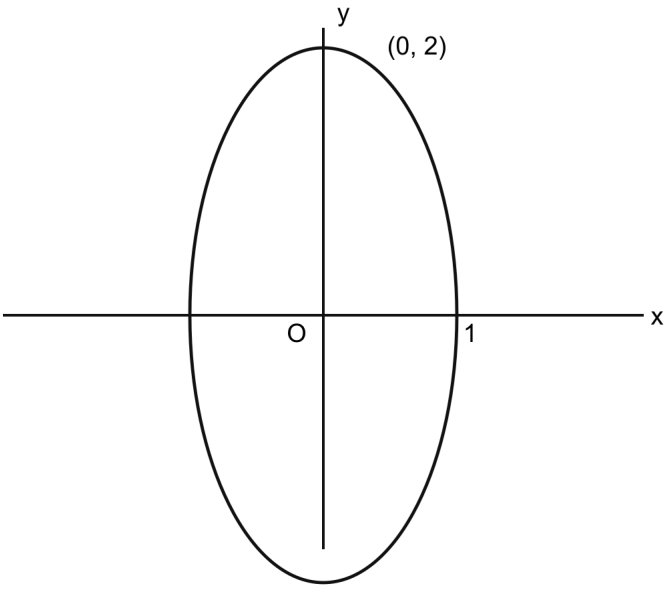
Non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Prove that $3\mathbf{a} - \mathbf{b}$ is perpendicular to $\mathbf{a} + 3\mathbf{b}$ if and only if $|\mathbf{a}| = |\mathbf{b}|$. [6]

SECTION A

Question 9 [Maximum mark: 9]

The ellipse C has equation $4x^2 + y^2 = 4$.



(a) Find the equation of the tangent to C at the point where $y = 2$. [4]

(b) The line $x\cos\theta + y\sin\theta = \sqrt{2}$ is tangent to C, where $0 < \theta < \pi/2$. Show that $\sin^2\theta = \frac{1}{3}$. [5]

SECTION B

Question 10 [Maximum mark: 18]

The continuous random variable X has probability density $f(x) = \frac{k}{1+x^2}$ for $0 \leq x \leq 1$, and $f(x) = 0$ otherwise, where $k > 0$.

(a) Find the exact value of k. [3]

(b) Write down $\mathrm{P}(X = 1/\sqrt{3})$. [1]

(c) Find the exact value of $\mathrm{P}(X \leq 1/\sqrt{3})$. [2]

(d) Show that $\mathrm{E}(X) = \frac{\ln b}{\pi}$ and find b. [4]

(e) Find the exact value of $\mathrm{E}(X^2)$. [5]

(f) The median is $M = \tan\theta$, where $0 < \theta < \pi/2$. Find the exact value of θ. [3]

SECTION B

Question 11 [Maximum mark: 17]

Let $f(x) = \dfrac{2e^{2x} - 1}{e^{2x} + 1}$, $x \in \mathbb{R}$.

(a) Find the coordinates of both intercepts of $y = f(x)$. [3]

(b) Show that f is strictly increasing. [4]

(c) Find $\lim_{x \to +\infty} f(x)$, justifying your answer. [3]

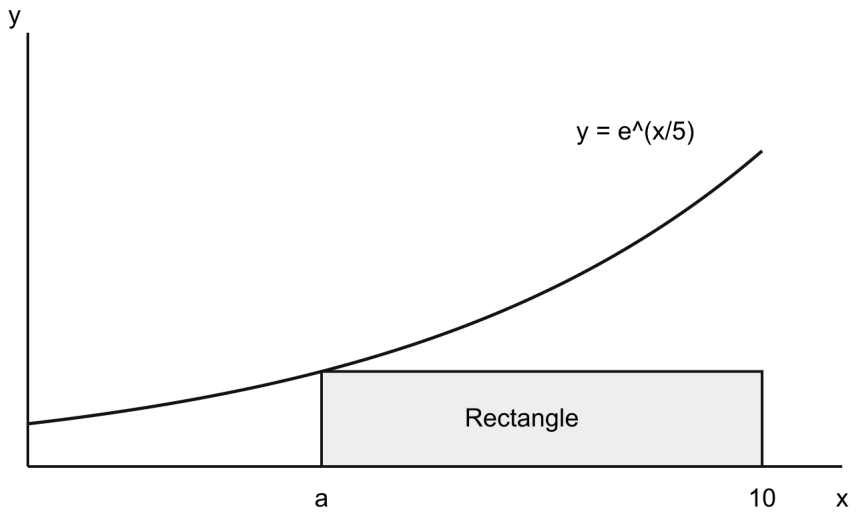
(d) Sketch $y = f(x)$, showing the intercepts and both horizontal asymptotes. [3]

(e) Explain why f^{-1} exists, and find the exact value of $f^{-1}(1)$. [4]

SECTION B

Question 12 [Maximum mark: 20]

Let $f(x) = e^{x/5}$. A rectangle has vertices $(a, 0)$, $(10, 0)$, $(10, f(a))$ and $(a, f(a))$, where $0 \leq a < 10$. Its area is S and perimeter is P.



(a) Find the equation of the tangent to $y = f(x)$ at $x = 0$. [3]

(b) The line $y = mx$ is tangent to $y = f(x)$. Find m. [5]

(c) Express S in terms of a. [2]

(d) Express P in terms of a. [2]

(e) Find $\mathrm{d}S/\mathrm{d}a$ in simplified form. [3]

(f) Find the exact perimeter when S is a maximum, justifying that the stationary value is a maximum. [5]

End of question paper

MATHEMATICS

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Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 5]

(a) Range: $y \neq -1$ **A1**

[1 mark]

(b) $f(x) = y \Leftrightarrow \frac{x-1}{2-x} = y$ **M1**

$\Leftrightarrow x - 1 = y(2 - x)$

$\Leftrightarrow x - 1 = 2y - xy$

$\Leftrightarrow x(1 + y) = 2y + 1$ **M1**

$\Leftrightarrow x = \frac{2y+1}{y+1}$ **A1**

Hence, $\Leftrightarrow f^{-1}(x) = \frac{2x+1}{x+1}$ **A1**

[4 marks]

2. [Maximum mark: 6]

$\int_0^{\frac{\pi}{2}} (8\sin x\cos x)\,dx = \int_0^{\frac{\pi}{2}} (4\sin 2x)\,dx$ **M1A1**

$= \left[-2\cos 2x\right]_0^{\pi/2}$ **A1**

$= -2\cos\pi + 2\cos 0$ **M1**

$= 2 + 2$ **A1**

$= 4$ **A1**

[6 marks]

3. [Maximum mark: 6]

(a) $S = \frac{-2a}{a} = -2$, $P = \frac{-3a}{a} = -3$ **A1 A1**

[2 marks]

(b) $f'(x) = 3ax^2 + 4ax + b$ **M1**

$f(1) = 3 \Leftrightarrow a + 2a + b + 3a = 3 \Leftrightarrow 6a + b = 3$ **A1**

$f'(1) = 0 \Leftrightarrow 3a + 4a + b = 0 \Leftrightarrow 7a + b = 0$ **A1**

$a = -3$, $b = 21$ **A1**

[4 marks]

4. [Maximum mark: 5]

(a) $p+0.7-p+q=1 \Leftrightarrow 0.7+q=1 \Leftrightarrow q=0.3$ **M1 A1**

[2 marks]

(b) $\mathrm{E}(X)=2.1 \Leftrightarrow p+2(0.7-p)+3q=2.1$ **M1 A1**

$\Leftrightarrow p+1.4-2p+0.9=2.1 \Leftrightarrow p=0.2$ **A1**

[3 marks]

5. [Maximum mark: 6]

Change of base: $\log_8(x+2)=\frac{\log_2(x+2)}{\log_2 8}=\frac{\log_2(x+2)}{3}$ **M1 A1**

Therefore,

$3\log_8(x+2)-2\log_2(x+1)=1$

$\Leftrightarrow \log_2(x+2)-2\log_2(x+1)=1$

$\Leftrightarrow \log_2\frac{(x+2)}{(x+1)^2}=1$ **M1**

$\Leftrightarrow \frac{(x+2)}{(x+1)^2}=2$ **A1**

$\Leftrightarrow x+2=2(x+1)^2 \Leftrightarrow x+2=2(x^2+2x+1)$

$\Leftrightarrow 2x^2+3x=0 \Leftrightarrow x(2x+3)=0$ **A1**

$\Leftrightarrow x=0$ or $x=-\frac{3}{2}$ (rejected)

Hence, $x=0$. **A1**

[6 marks]

6. [Maximum mark: 6]

(a) $S_3 = S_8 \Leftrightarrow \frac{3}{2}(2u_1 + 2d) = \frac{8}{2}(2u_1 + 7d)$ **M1 A1 A1**

$\Leftrightarrow 3u_1 + 3d = 8u_1 + 28d$

$\Leftrightarrow 0 = 5u_1 + 25d \Leftrightarrow 5u_1 = -25d$ **A1**

$\Leftrightarrow u_1 = -5d$

[4 marks]

(b) $u_6 = u_1 + 5d = 0$. **M1 A1**

[2 marks]

7. [Maximum mark: 6]

$$\sin x + \cos x = \sqrt{\frac{3}{2}}$$

Take square on both sides (valid since both sides are positive in the first quadrant).

$$(\sin x + \cos x)^2 = \frac{3}{2}$$ **M1**

$$\Leftrightarrow \sin^2 x + 2\sin x \cos x + \cos^2 x = \frac{3}{2}$$

$$\Leftrightarrow 1 + \sin 2x = \frac{3}{2}$$ **A1 A1**

$$\Leftrightarrow \sin 2x = \frac{1}{2}$$ **A1**

$2x = 30°,\ 2x = 150°$

$x = 15°,\ x = 75°$ **A1 A1**

[6 marks]

8. [Maximum mark: 6]

$\boldsymbol{a} \perp \boldsymbol{b} \Rightarrow \boldsymbol{a} \cdot \boldsymbol{b} = 0$ **A1**

$(3\boldsymbol{a} - \boldsymbol{b}) \cdot (\boldsymbol{a} + 3\boldsymbol{b}) = 3\boldsymbol{a}^2 + 9\boldsymbol{ab} - \boldsymbol{ab} - 3\boldsymbol{b}^2 = 3\boldsymbol{a}^2 - 3\boldsymbol{b}^2$

Hence,

$3\boldsymbol{a} - \boldsymbol{b} \perp \boldsymbol{a} + 3\boldsymbol{b} \Leftrightarrow (3\boldsymbol{a} - \boldsymbol{b}) \cdot (\boldsymbol{a} + 3\boldsymbol{b}) = 0$ **M1**

$\Leftrightarrow 3\boldsymbol{a}^2 - 3\boldsymbol{b}^2 = 0$

$\Leftrightarrow 3\boldsymbol{a}^2 = 3\boldsymbol{b}^2$

$\Leftrightarrow \boldsymbol{a}^2 = \boldsymbol{b}^2$ **A1**

$\Leftrightarrow |\boldsymbol{a}|^2 = |\boldsymbol{b}|^2$ **M1**

$\Leftrightarrow |\boldsymbol{a}| = |\boldsymbol{b}|$ **A1**

(**EITHER** $\Leftrightarrow$ must be seen **OR** two directions of proof must be seen) **M1**

[6 marks]

9. [Maximum mark: 9]

$4x^2 + y^2 = 4$.

(a) When $y = 2$, $x = 0$ **A1**

Implicit: $8x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{4x}{y}$ **A1**

$m_T = 0$ **A1**

Equation of tangent: $y = 2$. **A1**

[4 marks]

(b) $x\cos\theta + y\sin\theta = \sqrt{2} \Rightarrow y\sin\theta = \sqrt{2} - x\cos\theta \Rightarrow y = \sqrt{2}\,\text{cosec}\,\theta - x\cot\theta$

The two curves intersect:

$4x^2 + y^2 = 4 \Leftrightarrow 4x^2 + \left(\sqrt{2}\,\text{cosec}\,\theta - x\cot\theta\right)^2 = 4$ **M1**

$4x^2 + 2\,\text{cosec}^2\theta - 2\sqrt{2}x\,\text{cosec}\,\theta\cot\theta + x^2\cot^2\theta = 4$

$(4 + \cot^2\theta)x^2 - 2\sqrt{2}x\,\text{cosec}\,\theta\cot\theta + 2\,\text{cosec}^2\theta - 4 = 0$

(or any equivalent quadratic) **A1**

For tangency, the discriminant must be 0. **M1**

$\Delta = 8\,\text{cosec}^2\theta\cot^2\theta - 4(4 + \cot^2\theta)\left(2\,\text{cosec}^2\theta - 4\right) = 0$

$2\,\text{cosec}^2\theta\cot^2\theta = (4 + \cot^2\theta)\left(2\,\text{cosec}^2\theta - 4\right)$

$2\,\text{cosec}^2\theta\cot^2\theta = 8\,\text{cosec}^2\theta + 2\,\text{cosec}^2\theta\cot^2\theta - 16 - 4\cot^2\theta$

$0 = 8\,\text{cosec}^2\theta - 16 - 4\cot^2\theta$

$2\,\text{cosec}^2\theta = 4 + \cot^2\theta$ **A1**

$2\,\text{cosec}^2\theta = 4 + \text{cosec}^2\theta - 1$

$\text{cosec}^2\theta = 3$ **A1**

$\sin^2\theta = \frac{1}{3}$

[5 marks]

Friendly remark:

I wanted to share some thoughts regarding question 9(b). It does seem to be quite challenging and could prove to be more time-consuming than beneficial. As an alternative strategy, you might consider focusing on getting just 1 or 2 marks from this question and then (or even directly!) moving on to section B, where there are many opportunities to earn marks with relatively easier questions.

Section B

10. [Maximum mark: 18]

(a) $\int_0^1 f(x)\mathrm{d}x = 1 \Leftrightarrow \int_0^1 \frac{k}{1+x^2}\mathrm{d}x = 1$ **M1**

$\Leftrightarrow \left[k \arctan x\right]_0^1 = 1 \Leftrightarrow k\frac{\pi}{4} = 1$ **A1**

$\Leftrightarrow k = \frac{4}{\pi}$ **A1**

[3 marks]

(b) $\mathrm{P}\left(X = \frac{1}{\sqrt{3}}\right) = 0$ **A1**

[1 mark]

(c) $\mathrm{P}\left(X \leq \frac{1}{\sqrt{3}}\right) = \int_0^{1/\sqrt{3}} \frac{k}{1+x^2}\mathrm{d}x = \frac{4}{\pi}\left[\arctan x\right]_0^{1/\sqrt{3}} = \frac{4}{\pi}\arctan\frac{1}{\sqrt{3}} = \frac{4}{\pi}\times\frac{\pi}{6} = \frac{2}{3}$ **M1 A1**

[2 marks]

(d) $\mathrm{E}(X) = \frac{4}{\pi}\int_0^1 \frac{x}{1+x^2}\mathrm{d}x = \frac{2}{\pi}\left[\ln(1+x^2)\right]_0^1 = \frac{2}{\pi}\ln 2$ **M1 A1 A1**

$= \frac{\ln 4}{\pi}$, hence $b = 4$. **A1**

[4 marks]

(e) $\mathrm{E}(X^2) = \frac{4}{\pi}\int_0^1 \frac{x^2}{1+x^2}\mathrm{d}x$ **M1**

$= \frac{4}{\pi}\int_0^1 \frac{1+x^2-1}{1+x^2}\mathrm{d}x = \frac{4}{\pi}\int_0^1 1 - \frac{1}{1+x^2}\mathrm{d}x$ **M1 A1**

$= \frac{4}{\pi}\left[x - \arctan x\right]_0^1 = \frac{4}{\pi}\left(1 - \frac{\pi}{4}\right) = \frac{4}{\pi} - 1$ **A1 A1**

[5 marks]

(f) $\frac{4}{\pi}\int_0^{\mathrm{M}} \frac{1}{1+x^2}\mathrm{d}x = \frac{1}{2}$ **M1**

$\Leftrightarrow \frac{4}{\pi}\left[\arctan x\right]_0^{\mathrm{M}} = \frac{1}{2} \Leftrightarrow \frac{4}{\pi}\arctan \mathrm{M} = \frac{1}{2}$ **A1**

$\Leftrightarrow \arctan \mathrm{M} = \frac{\pi}{8} \Leftrightarrow \mathrm{M} = \tan\frac{\pi}{8}, \quad \theta = \frac{\pi}{8}$ **A1**

[3 marks]

11. [Maximum mark: 17]

$f(x)=\dfrac{2e^{2x}-1}{e^{2x}+1},\ x\in\mathbb{R}$.

(a) y-intercept: for $x=0$, $y=\dfrac{2-1}{1+1}=\dfrac{1}{2}$ **A1**

x-intercept: for $y=0$, $e^{2x}=\dfrac{1}{2}\Leftrightarrow 2x=\ln\dfrac{1}{2}\Leftrightarrow x=\dfrac{1}{2}\ln\dfrac{1}{2}$ (or $-\dfrac{1}{2}\ln 2$) **M1 A1**

[3 marks]

(b) $f'(x)=\dfrac{4e^{2x}(e^{2x}+1)-2e^{2x}(2e^{2x}-1)}{(e^{2x}+1)^2}$ **M1 A1**

$=\dfrac{4e^{4x}+4e^{2x}-4e^{4x}+2e^{2x}}{(e^{2x}+1)^2}$

$=\dfrac{6e^{2x}}{(e^{2x}+1)^2}>0$ **A1**

hence $f(x)$ is increasing. **A1**

[4 marks]

(c) **METHOD A**

$\lim\limits_{x\to+\infty}f(x)=\left(\dfrac{\infty}{\infty}\right)$ (L' Hospital) **R1**

$\lim\limits_{x\to+\infty}\dfrac{4e^{2x}}{2e^{2x}}=\lim\limits_{x\to+\infty}\dfrac{4}{2}=2$ **M1 A1**

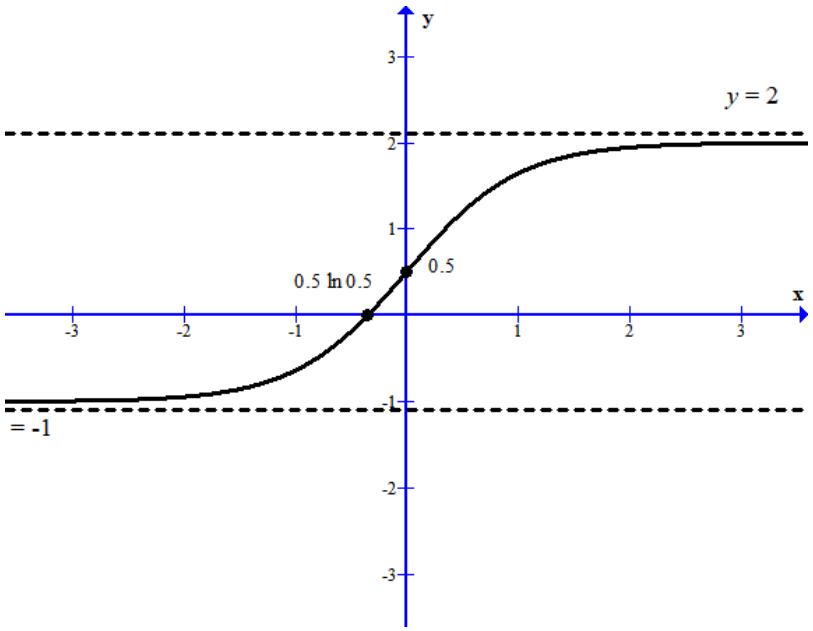
METHOD B

$\lim\limits_{x\to+\infty}f(x)=\lim\limits_{x\to+\infty}\dfrac{2-\dfrac{1}{e^{2x}}}{1+\dfrac{1}{e^{2x}}}=2$ **M1 A1**

since $\dfrac{1}{e^{2x}}$ tends to 0 **R1**

[3 marks]

(d)



A1 A1 A1

[3 marks]

(e) f^{-1} exists because f is 1-1 (increasing) **R1**

$f^{-1}(1) = x \Leftrightarrow f(x) = 1$ **M1**

$$\frac{2e^{2x} - 1}{e^{2x} + 1} = 1$$

$\Leftrightarrow 2e^{2x} - 1 = e^{2x} + 1 \Leftrightarrow e^{2x} = 2$ **A1**

$2x = \ln 2$

$x = \frac{1}{2}\ln 2, \quad f^{-1}(1) = \frac{1}{2}\ln 2$ **A1**

Note: You could find $f^{-1}(x)$ first and then $f^{-1}(1)$ but it is not necessary and takes more time!

[4 marks]

12. [Maximum mark: 20]

$f(x)=e^{\frac{x}{5}}$.

(a) At $x=0$, $y=1$.

$f'(x)=\frac{1}{5}e^{\frac{x}{5}}$ **M1 A1**

$m_T=\frac{1}{5}$

Tangent line: $y=\frac{1}{5}x+1$ **A1**

[3 marks]

(b) $f(x)=mx \Leftrightarrow e^{\frac{x}{5}}=mx$ **M1**

$f'(x)=m \Leftrightarrow \frac{1}{5}e^{\frac{x}{5}}=m$ **M1**

Hence, $e^{\frac{x}{5}}=\frac{1}{5}e^{\frac{x}{5}}x \Leftrightarrow 1=\frac{1}{5}x \Leftrightarrow x=5$ **A1 A1**

Hence, $m=\frac{e}{5}$ **A1**

[5 marks]

(c) The base is $10-a$ **A1**

The heigh is $f(a)=e^{\frac{a}{5}}$ **A1**

Hence $S=(10-a)e^{\frac{a}{5}}=10e^{\frac{a}{5}}-ae^{\frac{a}{5}}$

[2 marks]

(d) $P=2(10-a)+2e^{\frac{a}{5}}=20-2a+2e^{\frac{a}{5}}$ **A1 A1**

[2 marks]

(e) $\frac{\mathrm{d}S}{\mathrm{d}a} = 2\mathrm{e}^{\frac{a}{5}} - \left(\mathrm{e}^{\frac{a}{5}} + a\frac{1}{5}\mathrm{e}^{\frac{a}{5}}\right)$. **M1 A1**

$= \mathrm{e}^{\frac{a}{5}} - \frac{a}{5}\mathrm{e}^{\frac{a}{5}}$. **A1**

[3 marks]

(f) $\frac{\mathrm{d}S}{\mathrm{d}a} = 0 \Leftrightarrow \mathrm{e}^{\frac{a}{5}} = \frac{a}{5}\mathrm{e}^{\frac{a}{5}} \Leftrightarrow 1 = \frac{a}{5} \Leftrightarrow a = 5$ **M1 A1**

The second derivative is $\frac{\mathrm{d}^2S}{\mathrm{d}a^2} = \frac{1}{5}\mathrm{e}^{\frac{a}{5}} - \left(\frac{1}{5}\mathrm{e}^{\frac{a}{5}} + \frac{a}{25}\mathrm{e}^{\frac{a}{5}}\right) = -\frac{a}{25}\mathrm{e}^{\frac{a}{5}}$ **A1**

For $a = 5$, $\frac{\mathrm{d}^2S}{\mathrm{d}a^2} < 0$, hence local max. **R1**

Thus, $P = 20 - 10 + 2\mathrm{e} = 10 + 2\mathrm{e}$ **A1**

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[5 marks]